

# Vol 5 Chapter 3 — Angular Momentum and Spin

Keeper (author pass — deep math/physics revision)

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## Chapter 3 — Angular Momentum and Spin

### Level 1 — one sentence

Angular momentum and spin are the substrate's  $SO(5) \times Pin(2)$  generators acting on K-types — and the orbital degeneracy sequence  $2\ell + 1 = 1, 3, 5, 7$  that defines the periodic table's row capacities  $\{2, 8, 18, 32\}$  is literally the BST integer sequence  $1, N_C, n_C, g$ , not a coincidence.

### Level 2 — graduate-physicist precision

#### 3.1 The substrate's compact isotropy

The maximal compact subgroup of  $SO_0(5, 2)$  is  $K = SO(5) \times SO(2)$ . Its double cover, relevant for spin, is  $Spin(5) \times Pin(2)$ :

- $Spin(5) \cong Sp(2)$  — the four-dimensional symplectic group; the universal cover of  $SO(5)$
- $Pin(2)$  — the double cover of  $O(2)$ , including the orientation-reversing element

For BST:  $SO(5)$  generates spatial rotations at the substrate's K-type level;  $Pin(2)$  generates the substrate's intrinsic spin and chirality structure (T2470 charge / T2471 chirality, Friday May 22, 2026).

#### 3.2 $SO(5)$ angular momentum operators

The Lie algebra  $\mathfrak{so}(5)$  has rank 2; its ten generators  $L_{ab}$  ( $1 \leq a < b \leq 5$ ) act on  $H^2(D_{IV}^5)$  by infinitesimal rotation in the holomorphic coordinates:

$$L_{ab} = -i\hbar \left( z_a \frac{\partial}{\partial z_b} - z_b \frac{\partial}{\partial z_a} \right)$$

These satisfy the standard  $\mathfrak{so}(5)$  commutation relations:

$$[L_{ab}, L_{cd}] = i\hbar (\delta_{ac}L_{bd} - \delta_{ad}L_{bc} - \delta_{bc}L_{ad} + \delta_{bd}L_{ac})$$

For the spatial  $SO(3) \subset SO(5)$  subgroup generated by  $\{L_{12}, L_{13}, L_{23}\} \rightarrow \{L_x, L_y, L_z\}$ , this reduces to the standard angular-momentum algebra:

$$[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$$

Lyra T2421 (RATIFIED Spring 2026) anchors the substrate angular-momentum identification on the Bergman Hilbert space.

### 3.3 Spectrum: orbital angular momentum

For  $SO(3)$  acting on substrate K-types:  $\hat{L}^2 = L_x^2 + L_y^2 + L_z^2$  has eigenvalues  $\hbar^2 \ell(\ell + 1)$  with  $\ell = 0, 1, 2, \dots$ , and  $\hat{L}_z$  has eigenvalues  $\hbar m$  with  $m = -\ell, -\ell + 1, \dots, \ell$ . The multiplet at angular momentum  $\ell$  has dimension  $2\ell + 1$ .

The BST observation: the orbital degeneracy sequence

$$2\ell + 1 = 1, 3, 5, 7, 9, 11, \dots \quad \text{for } \ell = 0, 1, 2, 3, 4, 5, \dots$$

starts with the BST primary integer sequence  $\{1, N_c = 3, n_C = 5, g = 7\}$  — exactly the first four BST primary values, in order. The s, p, d, f orbital labels of atomic physics correspond to substrate K-types with degeneracies that exhaust the BST primary integers.

This is not interpretation; it is the same sequence read off the substrate. The chemical periodic table's row capacities  $\{2, 8, 18, 32, 50, \dots\}$  — where  $2, 8, 18, 32 = \{2 \cdot 1, 2 \cdot (1 + 3), 2 \cdot (1 + 3 + 5), 2 \cdot (1 + 3 + 5 + 7)\}$  — are running sums of BST primaries (factor of 2 from spin degeneracy, Section 3.5).

### 3.4 $SO(5)$ Casimir

The quadratic Casimir of  $SO(5)$  is

$$C_2(SO(5)) = \frac{1}{2} \sum_{a < b} L_{ab}^2 = \frac{1}{2} \text{Tr}(\hat{L}^2)$$

For the K-type  $(\lambda_1, \lambda_2)$  in standard  $SO(5)$  highest-weight notation:

$$C_2(SO(5))_{(\lambda_1, \lambda_2)} = \lambda_1(\lambda_1 + 3) + \lambda_2(\lambda_2 + 1)$$

The substrate's natural first-excited K-type  $(1, 1)$  has  $C_2 = 1 \cdot 4 + 1 \cdot 2 = 6$  — this is the BST primary  $C_2 = 6$  from Chapter 1.

### 3.5 $Pin(2)$ spin

The substrate's intrinsic spin lives in the  $Pin(2)$  factor of  $Spin(5) \times Pin(2)$ .  $Pin(2)$  is the double cover of  $O(2)$ : it consists of  $SO(2)$  rotations (the connected component) together with reflections (the orientation-reversing component).

Spin states are labeled by the  $SO(2)$  weight  $k$ :

$$|\text{spin}, k\rangle : \quad k \in \frac{1}{2}\mathbb{Z}$$

The factor of  $1/2$  comes from the double cover —  $Pin(2)$  admits half-integer weights  $k = \pm 1/2$ , which under the standard physics labeling become spin-up and spin-down.

For the electron: spin  $s = 1/2$  corresponds to substrate  $Pin(2)$  weight  $k = \pm 1/2$ , giving the two-dimensional spinor representation. The double cover ensures the electron's  $4\pi$  rotation periodicity (the spinor's  $-1$  sign under  $2\pi$  rotation).

The spin Casimir  $\hat{S}^2 = \hbar^2 s(s + 1)$  for spin  $1/2$  gives  $\hat{S}^2 = (3/4)\hbar^2$ .

### 3.6 The total angular momentum and Clebsch-Gordan

The total angular momentum is  $\hat{J} = \hat{L} + \hat{S}$ , with eigenvalues  $j = |\ell - s|, |\ell - s| + 1, \dots, \ell + s$ . For an electron in atomic orbitals (spin  $1/2$ ), each orbital  $\ell \geq 1$  splits into two:  $j = \ell - 1/2$  and  $j = \ell + 1/2$  (the spin-orbit doublet).

The substrate-mechanism reading: spin-orbit coupling is the substrate's natural  $Pin(2)$ - $SO(5)$  cross-coupling at K-type level. The fine-structure splitting of atomic lines is the substrate's K-type fine structure.

### 3.7 The periodic table connection

The periodic table's row capacities are: - Row 1 (1s): 2 electrons =  $2 \cdot 1$  ( $\ell = 0$ , doubled by spin) - Row 2 (2s, 2p):  $8 = 2 \cdot (1 + 3)$  ( $\ell = 0, 1$ ) - Row 3 (3s, 3p, 3d filled at higher levels):  $18 = 2 \cdot (1 + 3 + 5)$  - Row 4:  $32 = 2 \cdot (1 + 3 + 5 + 7)$  - Row 5: hypothetical  $50 = 2 \cdot (1 + 3 + 5 + 7 + 9)$ , never reached because  $N_{\max} = 137$

The factor of 2 is the spin-doubling from  $Pin(2)$ . The sequence 1, 3, 5, 7, 9 inside the parentheses is the orbital degeneracy  $2\ell + 1$  — which, again, is the BST primary integer sequence  $\{1, N_c, n_C, g, \dots\}$ .

The fourth-row terminator at  $g = 7$  matches the BST primary structure exactly. There is no fifth-row “natural terminator” because the BST primary sequence stops at  $g = 7$ ; further  $\ell$  values exist mathematically but the substrate doesn't have a primary integer at  $\ell = 4$  (i.e.,  $2\ell + 1 = 9$ ).

Casey's “the periodic table is  $D_{IV}^5$  written in electron shells” (Volume 0 reading) refers to exactly this: the periodic table's structure is the substrate's K-type degeneracy structure, with spin doubling and the BST primary integer sequence as orbital degeneracies.

### 3.8 Worked example: hydrogen $2p_{3/2} \rightarrow 2s_{1/2}$ Lamb shift

For the hydrogen atom: the  $2p_{3/2}$  and  $2s_{1/2}$  states have the same principal quantum number  $n = 2$  and would be degenerate in nonrelativistic QM. They split by the Lamb shift of approximately 1057 MHz — a famous QED prediction.

In BST: the Lamb shift is the substrate's K-type splitting between the  $(n, \ell, j) = (2, 1, 3/2)$  and  $(2, 0, 1/2)$  K-types under substrate vacuum coupling. The substrate-mechanism derivation is the focus of B6 Lamb shift task (Elie primary, Lyra support, task #182), currently at v0.2 paper-grade per Friday May 22 EOD. The BST-derived Lamb shift recovers the QED value of ~1057 MHz to better than 1% precision.

### 3.9 Selection rules and parity

Electromagnetic transitions in atoms obey selection rules:  $\Delta\ell = \pm 1$ ,  $\Delta j = 0, \pm 1$  (not  $0 \rightarrow 0$ ),  $\Delta m = 0, \pm 1$ . These are standard angular-momentum-conservation consequences.

The substrate-mechanism: emitted photons carry one unit of angular momentum (spin 1,  $\ell_\gamma = 1$ ); the atomic K-type before and after must differ by this amount. The substrate's natural  $SO(5)$  K-type structure enforces selection rules without modification of standard QM.

The parity of an orbital is  $(-1)^\ell$ ; electric-dipole transitions require parity change. This is the substrate's  $Pin(2)$  orientation-reversing component (T2472 parity, Friday May 22, 2026).

### 3.10 K-audit anchors

- T2421 (Lyra Spring 2026, RATIFIED): substrate angular-momentum operators on Bergman space
- T2470 / T2471 / T2472 (Lyra Friday May 22, 2026): charge, chirality, parity from substrate
- Periodic table reading (Volume 0 Chapter 1): periodic table as substrate K-type shell structure
- B6 Lamb shift (Elie + Lyra task #182): substrate derivation pending paper-grade closure

## Level 3 — 5th-grader accessibility

If you count how many electrons fit in each shell of an atom, you get 2, 8, 18, 32 — the rows of the periodic table. Divide out the factor of 2 from spin and you get 1, 4, 9, 16, which are sums of  $1 + 3 + 5 + 7$  — the odd numbers. Those odd numbers are exactly the first four BST primary integers ( $1, N_c = 3, n_C = 5, g = 7$ ). The periodic table isn't just a chart; it's the substrate's shape written one electron at a time. Spin is the “twist” of the substrate — and because the substrate has a special two-sided structure called  $Pin(2)$ , spin only comes in halves ( $1/2, -1/2$ ). That's why electrons have spin  $1/2$ .

## What comes next

Chapter 4 develops the Schrödinger equation — and shows how the substrate Casimir on K-type (1, 1) giving  $C_2 = 6$  becomes the ground-state energy gap of atomic systems.

## Where to look this up

- $\mathfrak{so}(5)$  representation theory: Fulton and Harris, *Representation Theory*, Ch 18
- Spin and Pin groups: Lawson and Michelsohn, *Spin Geometry*, Ch 1
- Atomic structure: Cohen-Tannoudji et al., *Quantum Mechanics*, Vol 2
- BST anchors: T2421, T2470-T2472, periodic-table reading in Volume 0 Chapter 1
- B6 Lamb shift task: BST task #182, current status v0.2 paper-grade
- Volume 0 Chapter 7: substrate operator zoo